1.20 Inequalities

Chapter 3 · Original slide 17 · Set Builder Notation

Recall that a set is a collection of values such that:

a) The order of values does not matter.

b) There are no duplicates.

Recap from GCSE:

• We use curly braces to list the values in a set, e.g. $A = \{1, 4, 6, 7\}$

• If A and B are sets then $A \cap B$ is the intersection of A and B, giving a set which has the elements in A and B.

• $A \cup B$ is the union of A and B, giving a set which has the elements in A or in B.

• $\varnothing$ is the empty set, i.e. the set with nothing in it.

• Sets can also be infinitely large. $\mathbb{N}$ is the set of natural numbers (all positive integers), $\mathbb{Z}$ is the set of all integers (including negative numbers and 0) and $\mathbb{R}$ is the set of all real numbers (including all possible decimals).

• We write $x \in A$ to mean "x is a member of the set A". So $x \in \mathbb{R}$ would mean "x is a real number".

$\{1, 2, 3\} \cap \{3, 4, 5\} =$

$\{1, 2, 3\} \cup \{3, 4, 5\} =$

$\{1, 2\} \cap \{3, 4\} =$

Chapter 3 · Original slide 18 · Set Builder Notation

It is possible to construct sets without having to explicitly list its values. We use:

The | or : means "such that".

$\{expr | condition\}$

or $\{expr : condition\}$

Can you guess what sets the following give?

$\{2x : x \in \mathbb{Z}\} =$

$\{2^x : x \in \mathbb{N}\} =$

$\{xy : x, y \text{ are prime}\} =$

We previously talked about 'solutions sets', so set builder notation is very useful for specifying the set of solutions!

1.20 Inequalities

Chapter 3 · Original slide 19 · Set Builder Notation

Can you use set builder notation to specify the following sets?

All (real) numbers greater than 5.

All (real) numbers between 5 and 7 inclusive.

All odd numbers.

All (real) numbers less than 5 or greater than 7.

Chapter 3 · Original slide 20 · Recap of linear inequalities

Inequality	Solution Set
$2x + 1 > 5$	
$-x \geq 2$	

$3(x - 5) \geq 5 - 2(x - 8)$

2 3 4

Note: Multiplying or both sides of an inequality by a negative number reverses the direction.

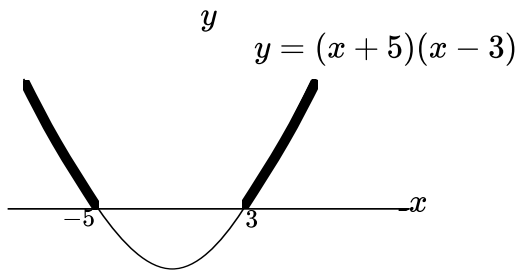
If $x < 3$ and $2 \leq x < 4$, what is the combined solution set?

Combining Inequalities:

If both inequalities have to be satisfied, we have to be on both lines. Place your finger vertically and scan across.

1.20 Inequalities

Chapter 3 · Original slide 21 · RECAP :: Solving Quadratic Inequalities



Solve $x^2 + 2x - 15 > 0$

Step 1: Get 0 on one side

$(x+5)(x-3) > 0$

Step 2: Factorise

Step 3: Sketch and reason

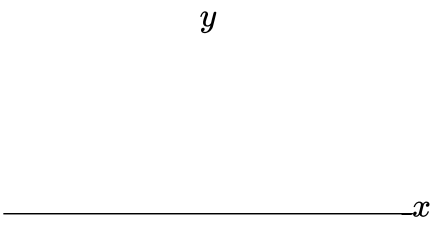
Since we sketched $y = (x+5)(x-3)$
we're interested where $y > 0$, i.e. the parts of the line where the y value is positive.

Click to bolden >

What can you say about the x values of points in this region?

What can you say about the x values of points in this region?

Chapter 3 · Original slide 22 · Solving Quadratic Inequalities



Solve $x^2 + 2x - 15 \leq 0$

Step 1: Get 0 on one side

$(x+5)(x-3) \leq 0$

Step 2: Factorise

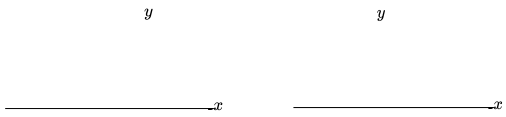
Step 3: Sketch and reason

Again, what can we say about the x value of any point in this region?

Note: we need $\leq$ rather than $<$ to be consistent with the original inequality.

1.20 Inequalities

Chapter 3 · Original slide 23 · Further Examples



Solve $x^2 + 5x \geq -4$

Solve $x^2 < 9$

Note: do not skip the 'sketch step'.

To use the quadratic inequality solver on ClassWiz, go to Menu → Inequalities, then choose 'order 2' (i.e. quadratic)

Try the same on a CG50.

Chapter 3 · Original slide 24 · Test Your Understanding

Find the set of values of x for which

(a) $3(x-2) < 8 - 2x$, (2)

(b) $(2x-7)(1+x) < 0$, (3)

(c) both $3(x-2) < 8 - 2x$ and $(2x-7)(1+x) < 0$. (1)

Note: the original equation has no solutions, but the inequality $q^2 + 8q < 0$ *did* have solutions. But we have found the solutions for q that result in the original equation not having any solutions for x. These are different variables, so have different solutions sets, even if the solution set of q influences the solution set of x.

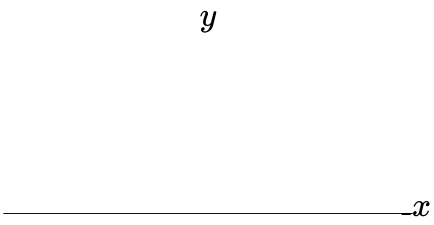
Given that the equation $2qx^2 + qx - 1 = 0$, where q is a constant, has no real roots,

(a) show that $q^2 + 8q < 0$. (2)

(b) Hence find the set of possible values of q. (3)

1.20 Inequalities

Chapter 3 · Original slide 25 · Deal with inequalities with a division by x



Find the set of values for which $\frac{6}{x} > 2, x \neq 0$

Why can't we just multiply both sides by x?

Chapter 3 · Original slide 26 · Exercise 3D/3E – Y1 Page 47-48, 50-51

3 Use set notation to describe the set of values of x for which:

g $4(3x + 7) \leq 20$ or $2(3x - 5) \geq \frac{7-6x}{2}$

6 Find the set of values of x for which $x^2 - 5x - 14 > 0$.

1.20 Inequalities

Chapter 3 · Original slide 27 · Exercise 3D/3E – Y1 Page 47-48, 50-51

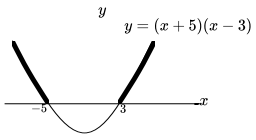
8 Given that $x \neq 3$, find the set of values for which $\frac{5}{x-3} < 2$.

Chapter 3 · Original slide 28 · Exercise 3D/3E – Y1 Page 47-48, 50-51

9 The equation $kx^2 - 2kx + 3 = 0$, where k is a constant, has no real roots. Prove that k satisfies the inequality $0 \leq k < 3$.

1.20 Inequalities

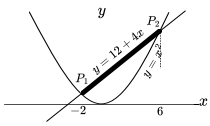
Chapter 3 · Original slide 29 · Inequalities on Graphs



When we solved quadratic inequalities, e.g. $(x+5)(x-3)>0$

We plotted $y=(x+5)(x-3)$ and observed the values of x for which $y>0$.

Can we use a similar method when we don't have 0 on one side?



Example: L_1 has equation $y=12+4x$. L_2 has equation $y=x^2$.

The diagram shows a sketch of L_1 and L_2 on the same axes.

a) Find the coordinates of P_1 and P_2, the points of intersection.

b) Hence write down the solution to the inequality $12+4x>x^2$.

a b

Solve simultaneously to find points of intersection: